Value-at-Risk (VaR) Analysis of a Multi-Asset Portfolio

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Abstract

This study evaluates portfolio risk using Value-at-Risk (VaR) techniques applied to a diversified portfolio of Indian equities. Historical simulation and multiple parametric models (Normal, Modified, EWMA, and Student-t) are implemented and compared. The analysis is extended to crisis periods and Monte Carlo simulation to study the effect of portfolio composition on risk. The results show that models incorporating fat tails and time-varying volatility provide more realistic risk estimates, especially during periods of market stress.

1 Introduction

Risk measurement is a fundamental aspect of financial decision-making. Value-at-Risk (VaR) is a widely used metric that estimates the maximum expected loss of a portfolio over a given time horizon at a specified confidence level.

For example, a 95% one-day VaR of 1.5% implies that there is only a 5% probability that the portfolio will lose more than 1.5% in a single day. However, financial returns are not normally distributed in reality. They exhibit skewness, heavy tails, and volatility clustering. Therefore, relying on a single VaR model may lead to inaccurate conclusions.

2 Objectives

- Construct a diversified portfolio
- Compute VaR using historical simulation
- Compare with parametric VaR models
- Analyze risk during crisis periods
- Study portfolio optimization using Monte Carlo simulation

3 Data and Portfolio Construction

The dataset consists of daily return data for ten major companies representing different sectors of the Indian economy, ensuring diversification. These include Reliance Industries and ONGC (energy), TCS and Infosys (information technology), HDFC Bank (banking), Bharti Airtel (telecommunications), ITC (consumer goods), Sun Pharma (pharmaceuticals), Tata Motors (automobile), and DMart (retail).

The inclusion of multiple sectors helps reduce unsystematic risk, as different industries respond differently to economic conditions. For instance, IT companies tend to be relatively stable, while energy and banking stocks are more sensitive to global and macroeconomic factors.

Before analysis, the dataset is preprocessed by converting the date column into datetime format, removing missing values, and selecting only relevant return columns. This ensures consistency and reliability of the data.

The portfolio is constructed using an equal-weighting approach, where each asset is assigned the same weight. This avoids bias toward any specific stock and ensures uniform contribution to portfolio performance.

The portfolio return is computed as:

$$R_p = \sum_{i=1}^{N} W_i R_i \tag{1}$$

Under equal weighting:

$$w_i = \frac{1}{N} \tag{2}$$

For a portfolio of 10 assets, each asset carries a weight of 10%. This approach allows diversification benefits, where losses in some assets can be offset by gains in others, making the portfolio more stable for risk analysis.

4 Methodology

4.1 Historical VaR

$$VaR_\alpha = Percentile_{(1-\alpha)}(R_p) \tag{3}$$

4.2 Parametric VaR Models

A single parametric VaR model is insufficient because financial markets do not follow a fixed distribution. Different models capture different characteristics such as symmetry, skewness, volatility clustering, and extreme events. Therefore, multiple models are used to obtain a comprehensive view of risk.

4.2.1 Normal VaR

$$VaR = \mu + z_\alpha \sigma \tag{4}$$

This is the simplest model and assumes normally distributed returns.
Why used: Provides a quick baseline estimate of risk.
Limitation: Underestimates extreme losses due to thin tails.

4.2.2 Modified VaR (Cornish-Fisher)

$$z_{cf} = z + \frac{1}{6}(z^2 - 1)S + \frac{1}{24}(z^3 - 3z)K - \frac{1}{36}(2z^3 - 5z)S^2 \tag{5}$$

This adjusts for skewness and kurtosis.
Why used: Corrects non-normality in returns.
Advantage: More realistic than Normal VaR.

4.2.3 EWMA VaR

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1-\lambda) r_t^2 \tag{6}$$

This models time-varying volatility.
Why used: Captures changing market risk over time.
Advantage: Reacts quickly to volatility changes.

4.2.4 Student-t VaR

This accounts for fat tails.

$$VaR = \mu + t_{\alpha,\nu} \cdot \sigma \cdot \sqrt{\frac{\nu - 2}{\nu}} \tag{7}$$

Why used: Models extreme losses more accurately.
Advantage: Best suited for crisis conditions.

Key Insight: Each model captures a different aspect of risk:

- Normal: average behavior
- Modified: skewness and kurtosis
- EWMA: time-varying volatility
- Student-t: extreme tail risk

4.3 Purpose of Monte Carlo Simulation

Monte Carlo simulation is used to analyze how portfolio weights affect risk by generating a large number of random portfolios. For each simulation, a weight vector $w = (w_1, w_2, ..., w_N)$ is generated such that:

$$\sum_{i=1}^{N} w_i = 1, \quad w_i \geq 0 \tag{8}$$

The portfolio return is computed as:

$$R_p = w^\top R \tag{9}$$

where R represents the vector of asset returns. The Value-at-Risk for each portfolio is then estimated using:

$$VaR_\alpha = Percentile_{(1-\alpha)}(R_p) \tag{10}$$

By repeating this process for many simulated portfolios, the distribution of VaR is obtained. This allows identification of the portfolio with minimum risk:

$$\min_{w} VaR_\alpha(w) \tag{11}$$

Thus, Monte Carlo simulation demonstrates how portfolio allocation influences risk and helps in finding the optimal portfolio with the lowest VaR.

5 Results and Discussion

5.1 Portfolio Characteristics

The portfolio consists of 10 equally weighted assets with the following statistical properties:

- Mean daily return: 0.0784%
- Standard deviation: 1.0881%
- Skewness: -0.8860
- Kurtosis: 13.9082

Interpretation: The negative skewness indicates a higher probability of extreme negative returns. The very high kurtosis confirms the presence of heavy tails, meaning extreme market movements occur more frequently than predicted by a normal distribution.

5.2 VaR Estimates (95%)

Insight: The Student-t VaR is the highest, reflecting its ability to capture extreme tail risk. The Normal VaR overestimates compared to historical in this case, while Modified VaR adjusts closer due to skewness and kurtosis.

5.3 Interpretation of Return Distribution

The return distribution plot with VaR thresholds provides a visual perspective of risk beyond numerical values.

- **Model Comparison:** VaR estimates are shown as vertical lines, making it easy to compare how conservative each model is. Student-t VaR lies further in the tail, indicating higher risk.
- **Tail Risk:** The left tail highlights the presence of extreme losses, showing that large negative returns occur more frequently than predicted by normal assumptions.
- **Non-Normality:** The deviation of the histogram from the normal curve indicates skewness and heavy tails in the data.
- **Better Intuition:** The plot shows where VaR lies within the distribution, giving a clearer understanding of how extreme the estimated loss actually is.

5.4 VaR at 99%

Historical VaR: 2.7261%
Insight: Higher confidence levels significantly increase VaR, highlighting sensitivity to extreme losses.

5.5 Why Not Emphasize 99% VaR Comparison

While VaR at the 99% confidence level is also computed, the primary comparison in this study is based on the 95% VaR.

- **Industry Standard:** The 95% VaR is widely used as a benchmark in financial analysis and provides a good balance between risk sensitivity and stability.
- **Statistical Reliability:** The 99% VaR focuses on extreme tail events, which depend on very few observations in historical data. This makes it less stable and more sensitive to outliers.
- **Model Comparison Clarity:** Differences between models are more consistently observable at the 95% level, making comparison more meaningful.
- **Complementary Role of 99% VaR:** The 99% VaR is still important as it captures extreme risk, but it is used as a supporting measure rather than the primary comparison metric.

5.6 Crisis Period Analysis

Key Insights:

- **Higher Risk During Crises:** VaR values increase significantly during crisis periods (e.g., COVID-19), indicating a sharp rise in potential losses compared to normal market conditions.

- **Underestimation in Normal Periods:** Risk models based on stable periods tend to underestimate losses, as they do not capture sudden spikes in volatility.
- **Importance of Tail Risk:** Extreme losses become more frequent during crises, highlighting the need for fat-tailed models like Student-t.
- **Model Sensitivity:** Differences between VaR models become more noticeable during crises, showing that model choice matters more in stressed markets.

5.7 Monte Carlo Analysis

Monte Carlo simulation (10,000 portfolios) is used to study the impact of asset allocation on risk.
Results:

- Minimum VaR: 1.2861%
- Mean VaR: 1.5482%
- Maximum VaR: 2.2644%

Insights:

- Minimum VaR is lower than equal-weighted VaR (1.4640%), showing optimization reduces risk.
- Most portfolios cluster around $\sim$ 1.55%, indicating typical risk levels.
- Poor weight choices can significantly increase VaR.

Optimal Weights: TCS (19.54%), HDFC (17.49%), ITC (16.97%), INFY (11.77%), DMART (10.22%), others smaller.

6 Limitations

- **Limited Asset Universe:** The study considers only ten stocks. A larger and more diverse set of assets could provide a more representative estimate of market risk.
- **Single Time Horizon:** VaR is computed for a one-day horizon only. Risk characteristics may differ for longer holding periods.
- **Historical Dependence:** The analysis relies on past return data. If future market conditions differ significantly, the estimates may not remain accurate.
- **Monte Carlo Simulation Based on Historical Returns:** The simulation generates random portfolio weights but still relies on historical data. It does not simulate new return paths or incorporate stochastic processes, limiting its ability to model unseen scenarios.
- **Absence of Extreme Loss Measure:** VaR only provides a threshold loss and does not quantify losses beyond that point. Hence, the magnitude of extreme losses is not captured.

7 Future Scope

- **Expected Shortfall (CVaR):** In addition to VaR, Conditional VaR can be implemented to measure the expected loss beyond the VaR threshold:

$$CVaR_\alpha = E[R_p | R_p \leq VaR_\alpha] \tag{12}$$

- **Full Monte Carlo Simulation of Returns:** Instead of using historical returns, stochastic models (e.g., Geometric Brownian Motion) can be used to simulate future return paths:

$$dS_t = \mu S_t dt + \sigma S_t dW_t \tag{13}$$

- **Portfolio Optimization Under Risk Constraints:** Optimization techniques can be applied to minimize VaR or CVaR subject to return constraints:

$$\min_w VaR_\alpha(w) \quad \text{s.t.} \quad E[R_p] \geq R^* \tag{14}$$

- **Dynamic Correlation Modeling:** Future work can incorporate time-varying correlations between assets, especially during crisis periods when correlations tend to increase.

8 Conclusion

This study demonstrates that portfolio risk cannot be accurately captured using a single Value-at-Risk model. While the Normal VaR provides a simple baseline, it fails to account for non-normal characteristics such as skewness, heavy tails, and volatility clustering observed in real financial data.

The empirical results show significant negative skewness and high kurtosis, confirming the presence of extreme events. As a result, models such as Modified VaR, EWMA, and especially Student-t VaR provide more realistic estimates by incorporating distribution asymmetry, time-varying volatility, and tail risk.

Crisis period analysis further highlights that risk increases sharply during stressed market conditions, and traditional models calibrated on normal periods tend to underestimate potential losses. This reinforces the importance of stress testing and model comparison in practical risk management.

Additionally, Monte Carlo simulation illustrates that portfolio construction plays a crucial role in risk reduction. The optimized portfolio achieves a lower VaR than the equal-weighted portfolio, demonstrating that appropriate asset allocation can significantly improve risk outcomes.

Overall, the study emphasizes that robust risk assessment requires a combination of models, consideration of extreme market conditions, and careful portfolio design. Such a multi-dimensional approach leads to more reliable and realistic estimation of financial risk.

9 References

- Hull, J. C., Risk Management and Financial Institutions
- `https://en.wikipedia.org/wiki/Value_at_risk`
- `https://www.investopedia.com/articles/04/092904.asp`

10 Live Application

An interactive implementation of the Portfolio Value-at-Risk (VaR) dashboard is deployed online and can be accessed via the following link: `https://financialproject-group18.streamlit.app/`